MTH 4322 – Fall 2026 – Project 1: A transportation system.

You work for a chain of stores that is spread across a wide geographic area. You have:

- A list of n_1 types of items that are sold at the stores.
- A list of n_2 warehouses, each with a list of how many items they currently have of each type.
- A list of n_3 stores, where each store has a list of stock that is currently missing. That is, for each item, it has an amount it needs to be shipped to it.
- A table of the shipping costs of a single item from each warehouse to each store. That is, "Shipping one unit of any item from warehouse X to store Y costs Z."

Write a program with a friendly user interface for the above situation. That is, the user can see the status of each store and warehouse, add required items for a store, add items to a warehouse, and so on.

There is also a button that finds the cheapest way of restocking all stores according to their current needs. This must be done efficiently, with respect to asymptotic running time in n_1, n_2, n_3, m, where m is the total number of items currently missing in the stores.

You make some decisions on your own. For example, think about what should happen if there is not enough inventory to satisfy all store needs. What else is missing?

Make sure you understand the asymptotic running time and the algorithm used.

You may assume that the stores, warehouses, and types of items never change. You may ***not*** use a linear programming algorithm (you don't need to know what that is – just make sure AI doesn't use it).

More advanced and optional: Have one program for the server, another for a store, and another for a warehouse. Store and warehouse programs communicate with the main server.